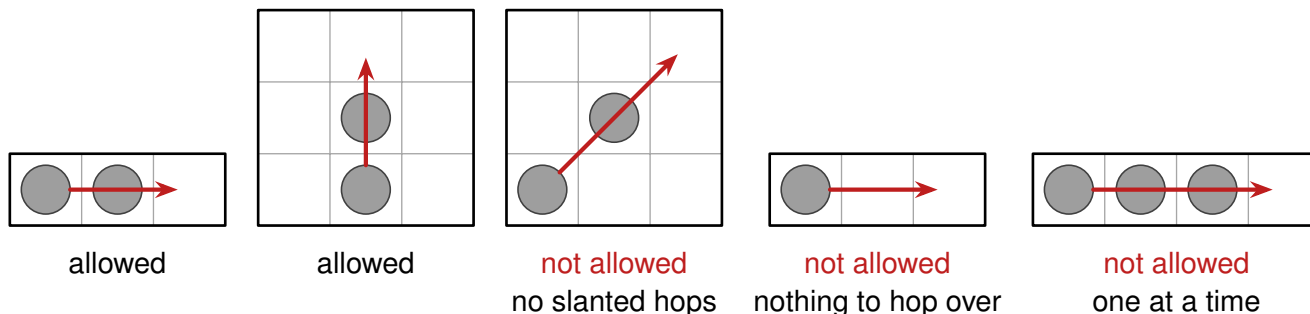


The Hop

The pennies are soldiers. A soldier may **hop over a neighbor** (up, down, left or right) into the **empty** square directly beyond. The soldier that was hopped over **goes home**: take it off the board. That is the only move in the game.



Warm-up: leave one

Put a penny on every gray circle. Hop until only **one** soldier is left. Write down how many hops it took. **One of the four is impossible.** Which one? How sure are you?

W1		hops: _____
W2		hops: _____
W3		hops: _____
W4		hops: _____

A counting rule

W4 starts with 4 soldiers. If you manage to leave one, how many hops did you make? Could a different order of hops take more hops, or fewer? Finish the sentence: *the number of hops is always* _____.

Across the River

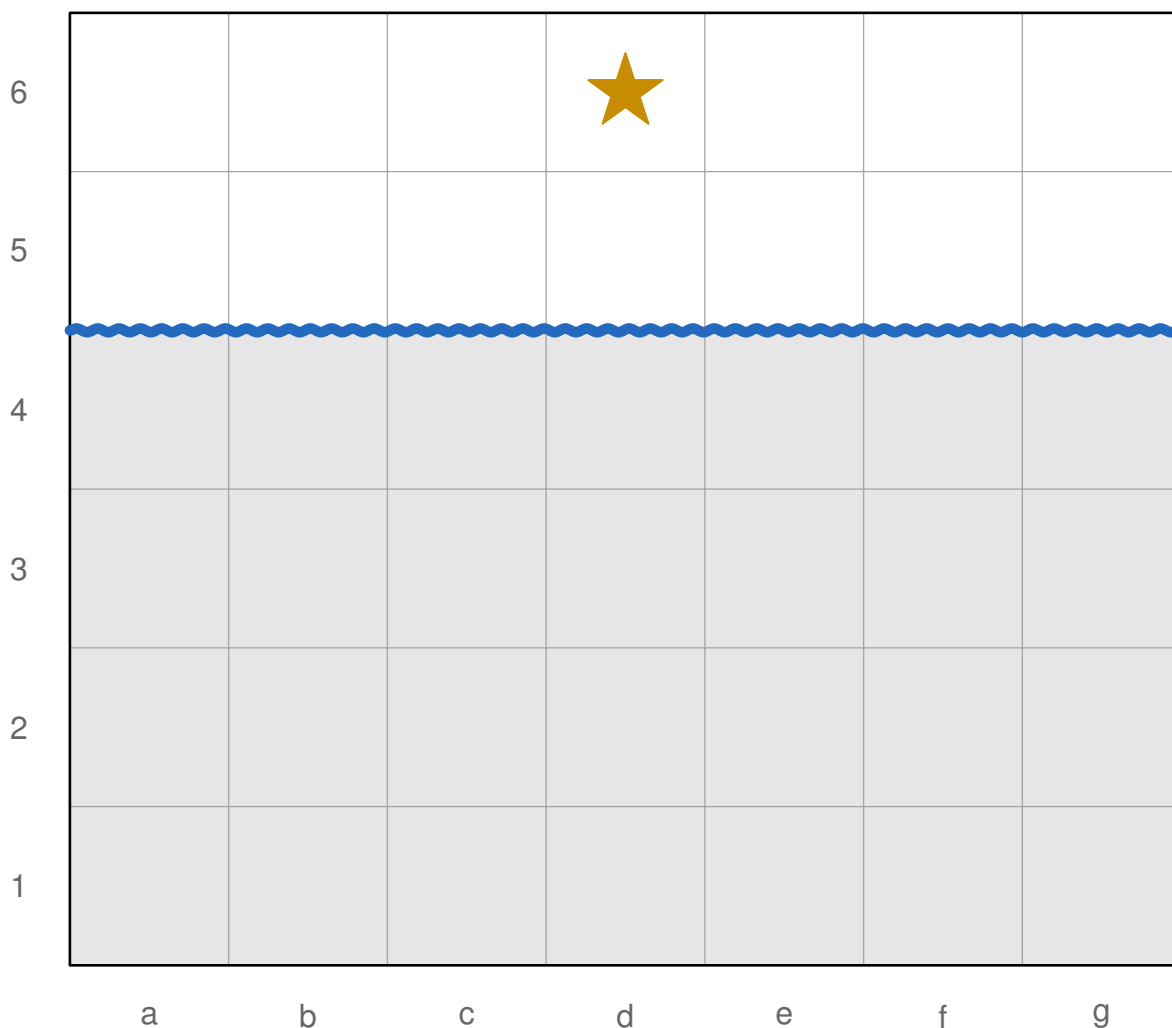
The gray squares are the army's camp; the blue line is the river; north of it every square is empty. **How far north can one soldier get?**

Mission 1: one row north

Two soldiers in a column, touching the river: the back one hops over the front one onto the far bank. Do it once.

Mission 2: two rows north

Get a soldier onto the star. Your soldiers may start on any gray squares. **How few soldiers can do it?** Try 3. Try 4.



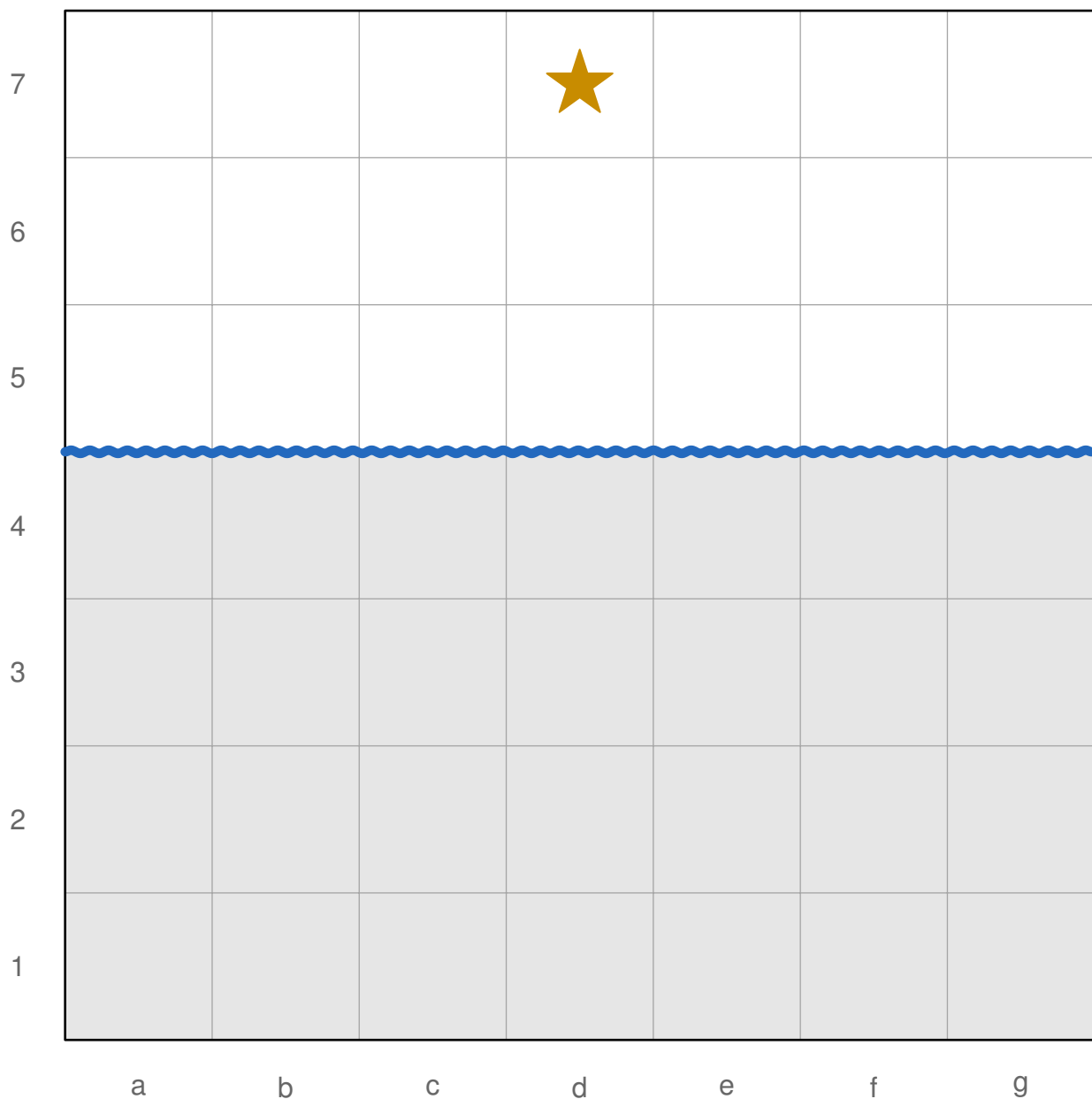
Name squares by column letter and row number, and write a hop as *from*→*to* (like d3→d5).

Army: _____ Hops: _____→_____→_____→_____→_____

Three Rows North

Mission 3: three rows north

Now the star is three rows past the river. Find an army that sends a soldier to the star. **It can be done with 8 soldiers.** Before you start: where must the *last* hop come from?



Army (list the squares): _____

Hops: _____ → _____ _____ → _____ _____ → _____ _____ → _____

_____ → _____ _____ → _____ _____ → _____ _____ → _____

Explorer's Page

The pattern

Fill in what you found. Then guess the next line — the true answer is a surprise, and your table leader knows it.

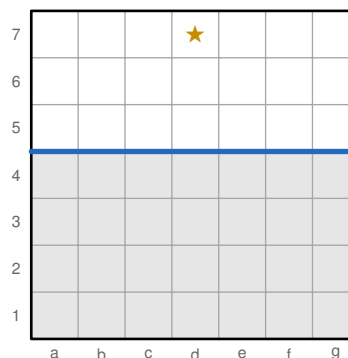
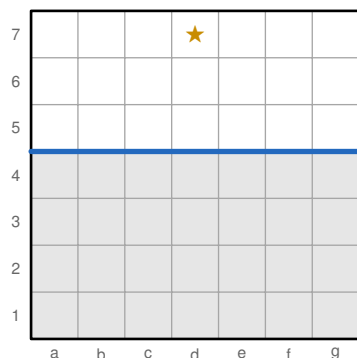
rows north of the river	fewest soldiers that can do it
1	
2	
3	
4	my guess:

Why not 3?

Somebody claims that 3 soldiers can get two rows north. Convince your table that it is impossible. Start like this: *the soldier that lands on the star must have hopped from _____ over _____, so just before the last hop there were soldiers on _____ and _____.* But...

Another army

There is more than one army of 8 that reaches three rows north. Find a second one, and draw both here (mark the river).



How many hops?

Your 8-soldier army sends one soldier to the star. Without redoing it, say how many hops it took. Does every 8-soldier solution take the same number of hops? Why?

Four rows north?

Ask your table leader for the army that reaches four rows north. It has 20 soldiers and takes 19 hops. Set it up on a big grid sheet — a puzzle to take home if we run out of time.

Scratch Board

Extra board for any mission. The river is between rows 4 and 5.

